



Biochar from Mulberry shoots: Turning Waste into Wealth



¹Anna Kaushik and ²Dr. P. Priyadharshini

¹*Research Scholar, Department of Sericulture, Forest College and Research Institute, Tamil Nadu Agricultural University, Mettupalayam - 641301, Tamil Nadu, India

²Assistant Professor, Department of Sericulture, Forest College and Research Institute, Tamil Nadu Agricultural University, Mettupalayam - 641301, Tamil Nadu, India

¹*Corresponding author (Email): anna.kaushik7@gmail.com

Abstract

Mulberry cultivation, a cornerstone of sericulture, produces large amounts of woody shoots during routine pruning, much of which remains underutilized. Converting this biomass into biochar through pyrolysis offers an effective pathway for waste valorisation, soil health improvement, and farm sustainability. Mulberry shoot-derived biochar is a stable, porous, carbon-rich material that enhances soil structure, water retention, nutrient availability, and microbial activity, leading to improved mulberry leaf yield and silkworm performance. Recent Indian studies highlight its benefits when co-applied with organic amendments, along with emerging roles in pollution remediation, carbon sequestration, and circular bioeconomy development.

Keywords: Mulberry, Sericulture, Biochar, Pyrolysis, Feedstock

Introduction

Mulberry cultivation forms the backbone of sericulture, yet it also generates a large volume of woody residues in the form of pruned shoots. Farmers often treat these shoots as waste, leaving them in heaps to decompose or burning them in the open, which not only adds to environmental pollution but also represents a lost resource. An alternative approach is to convert these residues into biochar, a stable, carbon-rich material created by heating biomass under limited oxygen. Biochar has gained attention worldwide for its ability to improve soil fertility, enhance water retention, and act as a long-term carbon sink. For mulberry growers, this means that what was once a disposal problem can be transformed into both a soil enhancer and a potential source of supplementary income. Importantly, field trials in India have already shown that mulberry-stalk biochar improves nutrient uptake and boosts leaf yield, directly supporting the health of silkworms and the quality of silk production (Nandini *et al.*, 2024; Lamani *et al.*, 2024). In this way, biochar bridges sustainable agriculture with circular waste management, offering sericulture farmers a practical solution that benefits both productivity and the environment.

What is Biochar?

Biochar is a carbon-rich material produced when plant residues or other organic matter are heated under conditions of limited oxygen, a process known as pyrolysis. Unlike simple ash or compost, biochar is highly porous and chemically stable. Its sponge-like structure helps soils retain water and nutrients, while its high surface area increases cation exchange capacity (CEC), meaning nutrients stay available to crops for longer. Another key feature is its persistence: once added to soil, biochar can remain for decades or even centuries without breaking down, making it a valuable tool for both improving soil health and locking away carbon that would otherwise be released into the atmosphere (Qu *et al.*, 2023). In this way, biochar functions as both a soil conditioner and a climate-friendly practice, offering benefits for farmers as well as the environment.

Why mulberry shoots make sense

Mulberry cultivation, which sustains India's silk industry, generates large amounts of lignocellulosic residues in the form of shoots and stalks after routine pruning. These by-products are often discarded as waste, yet they represent a readily available and renewable biomass source. In silk-producing regions such as Karnataka, Tamil Nadu, and Assam, the scale of mulberry farming ensures a steady and abundant supply of this material, making collection both practical and economical. From a chemical standpoint, mulberry shoots have a favourable composition rich in lignin and cellulose, which enhances the stability and porosity of biochar produced from them. This structural quality improves the soil's ability to hold water and nutrients. Field trials in India have reinforced these advantages, showing that biochar derived from local mulberry stalks can significantly boost nutrient availability in the soil, leading to higher mulberry leaf yields (Nandini *et al.*, 2024; Lamani *et al.*, 2024). Since leaf quality directly influences silkworm health and cocoon output, using shoots-derived biochar ties waste management directly to the productivity and profitability of sericulture itself. Figure 1 shows freshly pruned mulberry shoots, which are typically discarded but can instead be transformed into biochar for sustainable use.



Fig. 1: Mulberry shoots collected after pruning that serves as feedstock for biochar production (Source: Author's own photograph).

How a small farmer can make it (practical, low-cost)

For farmers, producing biochar doesn't always require expensive machinery or complex technology. The process can be done on a small scale using simple tools already available in many villages. The steps are straightforward:

Collect and prepare feedstock- After routine pruning, gather mulberry twigs, stalks, and shoots. Spread them out in the sun for several days to reduce moisture content. Dry wood burns more evenly and produces better-quality biochar (Lamani *et al.*, 2024).

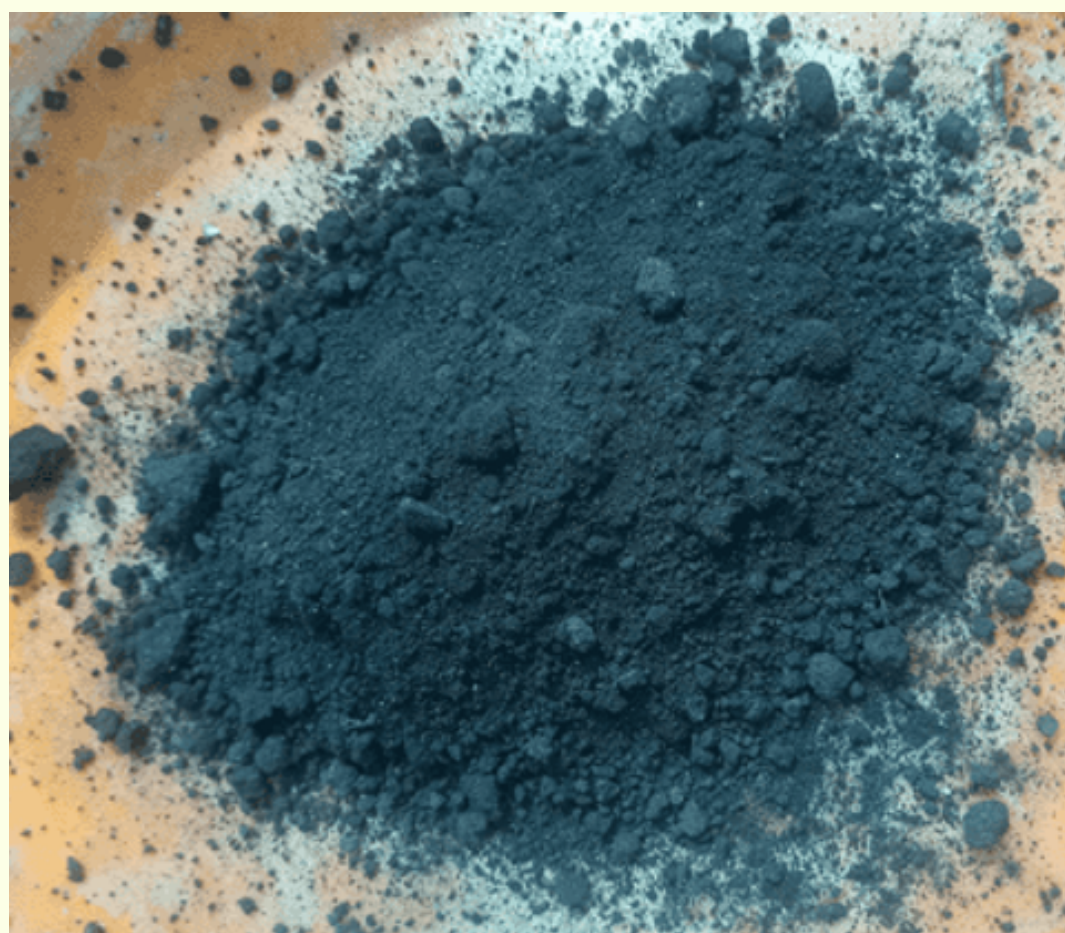
Build or use a basic kiln- A modified oil drum, a pit in the ground, or a small brick kiln can serve the purpose. The key is to maintain limited oxygen so the shoots undergo pyrolysis rather than open burning. Temperatures typically range from 300-600°C, which is sufficient to create stable biochar.

Cooling and crushing- Once the process is complete, seal the kiln and allow the charred shoots to cool. The cooled biochar can then be crushed or sieved into smaller granules, making it easier to mix with soil (Nandini *et al.*, 2024).

Application to fields- Biochar works best when blended with compost, vermicompost, or farmyard manure in ratios between 1:4 and 1:10. This combination enhances microbial activity and ensures nutrients are more readily available to crops (Lamani *et al.*, 2024).

Even simple, low-tech pyrolysis units can supply farmers with usable biochar, helping reduce mulberry waste while improving soil health. Although controlled kilns yield more uniform biochar, they are not essential for small-scale, on-farm use. This low-cost approach offers mulberry growers an accessible pathway to sustainable waste management.

As shown in **Figure 2**, the produced mulberry biochar is lightweight, porous, and suitable for blending with organic amendments before field application.



Key benefits

Soil health - Better structure, improved aeration, higher water retention, and more gradual nutrient release all mean less fertilizer use and stronger, more resilient mulberry plants. Indian field data confirm higher leaf yields and improved nutrient uptake when mulberry-stalk biochar was co-applied with farmyard manure (FYM) or NPK fertilizers (Nandini *et al.*, 2024; Lamani *et al.*, 2024).

Pollution control & remediation - Chemically modified mulberry-branch biochars have been tested as catalysts and adsorbents to remove organic pollutants like bisphenol A and reduce toxicity in lab bioassays. This shows promise for broader applications in wastewater treatment and contaminated soil stabilization (Qu *et al.*, 2023).

Heavy-metal cleanup - Engineered mulberry-stem biochar composites (for example, blended with hydroxyapatite) can strongly adsorb Pb^{2+} ions, making them valuable for cleaning soils or irrigation water contaminated with toxic heavy metals (Wang *et al.*, 2025).

Income & circularity - Instead of burning shoots, farmers can apply biochar to their own land or sell surplus as a local soil amendment. This reduces waste management costs, creates small income opportunities, and supports a circular economy approach in sericulture regions (Nandini *et al.*, 2024).

Climate resilience - Because biochar is highly stable, it locks carbon into the soil for decades to centuries, reducing greenhouse gas emissions. At the same time, its water-holding capacity helps crops tolerate dry spells, an important benefit in regions facing irregular rainfall (Qu *et al.*, 2023).

Energy co-benefits - Small-scale pyrolysis not only yields biochar but can also generate combustible gases and bio-oil, which can be tapped for cooking or heating, offering an added layer of rural energy security.

Improved nutrient-use efficiency - When used with chemical fertilizers, biochar reduces nutrient leaching, meaning more nitrogen and phosphorus stay in the soil for plant uptake rather than washing away, lowering costs and environmental impacts (Nandini *et al.*, 2024).

Simple Usage Tips (For Best Results)

Using biochar effectively is as important as making it. Farmers should remember that biochar is not a fertilizer by itself but a soil enhancer that works best when combined with organic matter. A few practical guidelines can help maximize its benefits:

Always mix with compost or manure - Blending biochar with compost or FYM before use “charges” the porous structure with nutrients and beneficial microbes. This step prevents biochar from drawing nutrients away from crops in the initial stages and ensures a more balanced soil amendment.

Apply at the right rate - Trials in India and abroad have shown good results when applying between 5-15 tonnes per hectare, with most studies recommending around 7.5-15 t/ha for visible improvements in soil fertility and mulberry leaf yield. Farmers with smaller budgets can start at the lower end of this range (Nandini *et al.*, 2024).

Experiment on small plots first - Since soil type and local conditions vary, it’s best to apply biochar on a small section of land while keeping a control plot without biochar. Comparing the two plots over one or two crop cycles helps farmers judge its real impact before scaling up across the whole farm.

Limitations and Things to Watch

Although biochar offers many advantages, farmers should be aware of certain limitations to avoid disappointment:

Nitrogen immobilization - Fresh, raw biochar tends to temporarily lock up nitrogen in the soil if applied alone, which may stunt plant growth. That’s why co-applying with compost or farmyard manure is strongly recommended, as it offsets this effect and supports healthy crop growth (Lamani *et al.*, 2024).

Quality variation - The properties of biochar depend on how it’s made. For example, low-temperature chars (below ~400°C) often contain more volatile matter and are less stable, breaking down faster in the soil. Higher-temperature chars are more stable but may lose some functional groups important for nutrient interaction. Small farmers should aim for a balance between practicality and quality, focusing on simple but controlled pyrolysis rather than open burning (Qu *et al.*, 2023; Wang *et al.*, 2025).

By keeping these points in mind, mulberry growers can get consistent benefits from shoots-based biochar while minimizing risks.

Conclusion

Turning mulberry shoots into biochar offers a practical way for sericulture farmers to manage waste while improving soil fertility, boosting leaf yield, and supporting healthier silkworm rearing. Beyond direct farm benefits, shoots-derived biochar also contributes to carbon sequestration, pollution control, and rural income opportunities, making it both a sustainability and livelihood solution. By adopting small-scale, low-cost production methods and using biochar wisely with compost or manure, farmers can close the loop between waste management and silk production, strengthening the resilience and productivity of sericulture systems.

References

Nandini, R., Shaziya, K. L., Hanumanta, D. L., & HC, P. (2024). Effect of Mulberry Stalk Biochar on Soil Nutrients Status and Productivity of Mulberry (*Morus alba* L.). *Asian Journal of Soil Science and Plant Nutrition*, 10(2), 382-395.

Lamani, H. D., KN, K. M., & Verma, R. (2024). Effect of Mulberry Stalk Biochar, FYM and Humic Acid on Growth, Yield and Quality of Mulberry (*Morus alba* L.). *International Journal of Plant & Soil Science*, 36(4), 174-183.

Qu, H., Chen, L., Yang, F., Zhu, J., Qi, C., & Peng, G. (2023). Synthesis of an environmentally friendly modified mulberry branch-derived biochar composite: high degradation efficiency of BPA and mitigation of toxicity in silkworm larvae. *International Journal of Molecular Sciences*, 24(4), 3609.

Wang, D., Zhou, X., Liang, M., & Wu, Z. (2025). Synthesis of Hydroxyapatite Mulberry Stem Biochar Composites for Efficient Pb (II) Adsorption from Aqueous Solutions. *Water*, 17(9), 1389.

